Exercise 9.9: Self-Generalizations of Hölder's Inequality

Why reciprocal exponents add when sequences are multiplied

A step-by-step tutorial for a high-school reader

What you will learn

By the end of this tutorial, you will be able to:

- read expressions such as $\left(\sum a_j^p\right)^{1/p}$;
- recognize a pair of *conjugate exponents*;
- turn the relation $1/p + 1/q = 1/r$ into the usual Hölder relation;
- prove the two-factor inequality in part (a) line by line;
- prove the three-factor inequality in part (b) by grouping two factors;
- understand the general rule that reciprocal exponents behave like a budget.

Contents

1 The exercise, stated carefully

The problem

Let $a_1, \dots, a_n$, $b_1, \dots, b_n$, and $c_1, \dots, c_n$ be nonnegative real numbers.
(a) Suppose $p, q, r > 0$, with $p > r$, $q > r$, and

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r}.$$

Prove that

$$\boxed{\left(\sum_{j=1}^{n} (a_j b_j)^r\right)^{1/r} \le \left(\sum_{j=1}^{n} a_j^p\right)^{1/p} \left(\sum_{j=1}^{n} b_j^q\right)^{1/q}.} \tag{A}$$

(b) Suppose $p, q, r > 1$ and

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1.$$

Prove that

$$\boxed{\sum_{j=1}^{n} a_j b_j c_j \le \left(\sum_{j=1}^{n} a_j^p\right)^{1/p} \left(\sum_{j=1}^{n} b_j^q\right)^{1/q} \left(\sum_{j=1}^{n} c_j^r\right)^{1/r}.} \tag{B}$$

What if the numbers can be negative?

If the entries are arbitrary real numbers, noninteger powers such as a_j^p may not even be real when $a_j < 0$. The standard real-valued versions use absolute values:

$$\left(\sum_{j=1}^{n} |a_j b_j|^r\right)^{1/r} \le \left(\sum_{j=1}^{n} |a_j|^p\right)^{1/p} \left(\sum_{j=1}^{n} |b_j|^q\right)^{1/q}, \tag{A$'$}$$

and

$$\left|\sum_{j=1}^{n} a_j b_j c_j\right| \le \left(\sum_{j=1}^{n} |a_j|^p\right)^{1/p} \left(\sum_{j=1}^{n} |b_j|^q\right)^{1/q} \left(\sum_{j=1}^{n} |c_j|^r\right)^{1/r}. \tag{B$'$}$$

We prove the nonnegative form printed in the exercise. Applying that proof to $|a_j|$, $|b_j|$, and $|c_j|$, together with the triangle inequality, gives the stronger real-valued forms above.

2 The small amount of background we need

2.1 Finite sequences and pointwise products

A finite sequence is simply an ordered list. For example,

$$a = (a_1, a_2, a_3) = (1, 2, 4), \qquad b = (b_1, b_2, b_3) = (3, 5, 2).$$

Their *pointwise product* is obtained by multiplying entries with the same index:

$$ab = (a_1 b_1, a_2 b_2, a_3 b_3) = (3, 10, 8).$$

The symbol j in a sum tells us which matching pair is being used:

$$\sum_{j=1}^{3} a_j b_j = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

2.2 Power-sum sizes

For a nonnegative sequence $a = (a_1, \ldots, a_n)$ and a positive number p, define

$$\|a\|_p := \left(\sum_{j=1}^{n} a_j^p \right)^{1/p}. \tag{1}$$

For example, if $a = (1, 2)$, then

$$\|a\|_1 = 1 + 2 = 3, \qquad \|a\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}, \qquad \|a\|_4 = (1^4 + 2^4)^{1/4} = 17^{1/4}.$$

When $p \geq 1$, this is called the ℓ^p-norm of the sequence. Part (a) allows $0 < r < 1$, in which case the same formula is still useful, although mathematicians do not technically call it a norm.

With this notation, part (a) is the compact statement

$$\|ab\|_r \leq \|a\|_p \, \|b\|_q \,. \tag{2}$$

2.3 Two exponent rules used repeatedly

For $x \geq 0$ and positive exponents, we use

$$(x^\alpha)^\beta = x^{\alpha\beta}. \tag{3}$$

Two instances will be especially important:

$$\left(a_j^r\right)^{p/r} = a_j^{r(p/r)} = a_j^p, \tag{4}$$

and, for $X \geq 0$,

$$\left(X^{r/p}\right)^{1/r} = X^{(r/p)(1/r)} = X^{1/p}. \tag{5}$$

The fractions p/r and r/p look similar, so it is worth slowing down whenever they appear.

2.4 Conjugate exponents

Conjugate exponents

Numbers $P, Q > 1$ are called *conjugate exponents* when

$$\frac{1}{P} + \frac{1}{Q} = 1. \tag{6}$$

Some common pairs are

P	Q	$1/P + 1/Q$
2	2	$1/2 + 1/2 = 1$
3	$3/2$	$1/3 + 2/3 = 1$
4	$4/3$	$1/4 + 3/4 = 1$

If one exponent is $P > 1$, its conjugate is

$$Q = \frac{P}{P-1}.$$

2.5 The ordinary two-factor Hölder inequality

The theorem that this exercise asks us to reuse is the following.

Ordinary Hölder inequality

Let $P, Q > 1$ satisfy $1/P + 1/Q = 1$. For nonnegative numbers x_j, y_j,

$$\boxed{\sum_{j=1}^{n} x_j y_j \le \left(\sum_{j=1}^{n} x_j^P\right)^{1/P} \left(\sum_{j=1}^{n} y_j^Q\right)^{1/Q}.} \tag{H}$$

When $P = Q = 2$, this is the familiar Cauchy–Schwarz inequality:

$$\sum_{j=1}^{n} x_j y_j \le \sqrt{\sum_{j=1}^{n} x_j^2}\sqrt{\sum_{j=1}^{n} y_j^2}.$$

The exercise does not ask us to reprove (H). Our task is to choose new sequences and new exponents so that (H) produces the desired formulas.

The strategy in one sentence

Do not force the original exponents p, q, r directly into Hölder's inequality. First *rescale the exponents* until their reciprocals add to 1.

3 The exponent-budget picture

Relations among reciprocal exponents are easier to remember if we imagine a bar whose total length is being divided into pieces.

3.1 Part (a): scale a total of $1/r$ into a total of 1

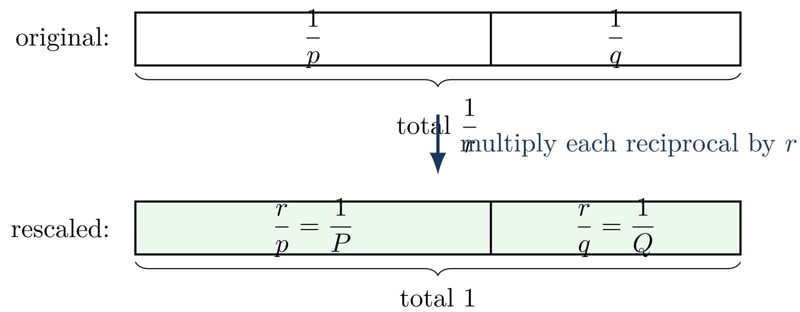
The hypothesis says

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r}.$$

Multiplying every term by r gives

$$\frac{r}{p} + \frac{r}{q} = 1. \tag{7}$$

This is exactly the conjugate-exponent pattern required by ordinary Hölder.



The correct new exponents are therefore

$$P = \frac{p}{r}, \qquad Q = \frac{q}{r}, \tag{8}$$

because then $1/P = r/p$ and $1/Q = r/q$.

3.2 Part (b): combine two pieces first

The hypothesis in part (b) divides a total reciprocal budget of 1 into three pieces:

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1.$$

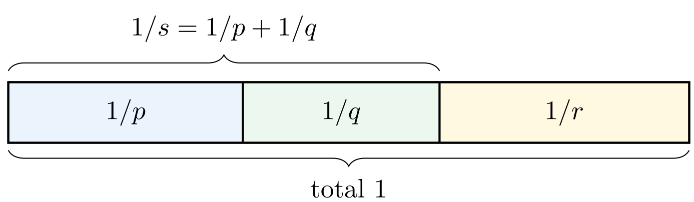
We will combine the first two pieces and call their total $1/s$:

$$\frac{1}{s} := \frac{1}{p} + \frac{1}{q}. \tag{9}$$

Then

$$\frac{1}{s} + \frac{1}{r} = 1, \tag{10}$$

so s and r are conjugate exponents.



This picture tells us the proof plan: first control the product $a_j b_j$ in the s-power sum, and then combine that product with c_j.

4 Part (a): the two-factor generalization

We assume throughout this section that

$$p, q, r > 0, \qquad p > r, \qquad q > r, \qquad \frac{1}{p} + \frac{1}{q} = \frac{1}{r}. \tag{11}$$

Step 1: manufacture conjugate exponents

Define

$$P := \frac{p}{r}, \qquad Q := \frac{q}{r}. \tag{12}$$

Because $p > r$ and $q > r$, we have $P > 1$ and $Q > 1$. Also,

$$\frac{1}{P} + \frac{1}{Q} = \frac{r}{p} + \frac{r}{q} = r\left(\frac{1}{p} + \frac{1}{q}\right) = r \cdot \frac{1}{r} = 1. \tag{13}$$

Thus P, Q are conjugate exponents, exactly as ordinary Hölder requires.

Step 2: choose the sequences to which Hölder will be applied

Set

$$x_j := a_j^r, \qquad y_j := b_j^r. \tag{14}$$

Then

$$x_j y_j = a_j^r b_j^r = (a_j b_j)^r. \tag{15}$$

This makes the left side of ordinary Hölder equal to the sum appearing inside the left side of (A).

Step 3: apply ordinary Hölder

Applying (H) to the sequences (x_j) and (y_j) with exponents P, Q gives

$$\begin{aligned} \sum_{j=1}^{n} (a_j b_j)^r &= \sum_{j=1}^{n} x_j y_j \\ &\le \left(\sum_{j=1}^{n} x_j^P\right)^{1/P} \left(\sum_{j=1}^{n} y_j^Q\right)^{1/Q}. \end{aligned} \tag{16}$$

Now simplify each power:

$$x_j^P = (a_j^r)^{p/r} = a_j^p, \qquad y_j^Q = (b_j^r)^{q/r} = b_j^q. \tag{17}$$

Also $1/P = r/p$ and $1/Q = r/q$. Therefore (16) becomes

$$\sum_{j=1}^{n} (a_j b_j)^r \le \left(\sum_{j=1}^{n} a_j^p\right)^{r/p} \left(\sum_{j=1}^{n} b_j^q\right)^{r/q}. \tag{18}$$

Step 4: take the positive r-th root

Every quantity in (18) is nonnegative. Since $r > 0$, the function $t \mapsto t^{1/r}$ is increasing for $t \geq 0$. Taking the r-th root preserves the inequality:

$$\begin{aligned}\left(\sum_{j=1}^{n}(a_jb_j)^r\right)^{1/r} &\leq \left[\left(\sum_{j=1}^{n}a_j^p\right)^{r/p}\left(\sum_{j=1}^{n}b_j^q\right)^{r/q}\right]^{1/r} \\ &= \left(\sum_{j=1}^{n}a_j^p\right)^{1/p}\left(\sum_{j=1}^{n}b_j^q\right)^{1/q}.\end{aligned} \tag{19}$$

This is exactly the desired inequality (A).

The whole proof compressed into one display

$$\begin{aligned}\sum_{j=1}^{n}(a_jb_j)^r &= \sum_{j=1}^{n}(a_j^r)(b_j^r) \\ &\leq \left(\sum_{j=1}^{n}(a_j^r)^{p/r}\right)^{r/p}\left(\sum_{j=1}^{n}(b_j^r)^{q/r}\right)^{r/q} \\ &= \left(\sum_{j=1}^{n}a_j^p\right)^{r/p}\left(\sum_{j=1}^{n}b_j^q\right)^{r/q}.\end{aligned}$$

The first line prepares the product; the second line is ordinary Hölder; the third line simplifies exponents. Taking the r-th root finishes the argument.

4.1 Why the condition $p, q > r$ appears

Ordinary Hölder needs exponents greater than 1. Our new exponents are $P = p/r$ and $Q = q/r$, so we need

$$\frac{p}{r} > 1, \qquad \frac{q}{r} > 1,$$

which is precisely $p > r$ and $q > r$.

In fact, once all three numbers are positive, the reciprocal equation already forces these inequalities. For example,

$$\frac{1}{r} = \frac{1}{p} + \frac{1}{q} > \frac{1}{p}.$$

For positive numbers, a larger reciprocal corresponds to a smaller number, so $r < p$. Similarly, $r < q$.

4.2 A concrete equality example

Take

$$p = q = 4, \qquad r = 2.$$

Indeed,

$$\frac{1}{4} + \frac{1}{4} = \frac{1}{2} = \frac{1}{r}.$$

Let $a = b = (1, 2)$. The left side is

$$\Big((1\cdot 1)^2 + (2\cdot 2)^2\Big)^{1/2} = \sqrt{1+16} = \sqrt{17}.$$

The right side is

$$(1^4 + 2^4)^{1/4}(1^4 + 2^4)^{1/4} = 17^{1/4}17^{1/4} = \sqrt{17}.$$

Thus equality can occur.

4.3 When does equality occur?

For nonzero nonnegative sequences, equality in ordinary Hölder occurs when the powered sequences are proportional. Here that condition becomes

$$(a_j^r)^P = \lambda(b_j^r)^Q \qquad \text{for every } j,$$

or equivalently

$$\boxed{a_j^p = \lambda b_j^q \quad \text{for every } j} \tag{20}$$

for some constant $\lambda > 0$. The example $p = q = 4$, $a = b$ satisfies this with $\lambda = 1$.

5 Part (b): the triple-product inequality

Now assume

$$p, q, r > 1, \qquad \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1. \tag{21}$$

The goal is to control a product of *three* sequences using the two-factor Hölder inequality.

Group two factors into one

Write

$$a_j b_j c_j = (a_j b_j) c_j.$$

We will first treat $a_j b_j$ as one sequence and pair it with c_j. Part (a) will then separate a_j and b_j again.

Step 1: define the intermediate exponent s

Let s be defined by

$$\frac{1}{s} := \frac{1}{p} + \frac{1}{q}. \tag{22}$$

Using (21),

$$\frac{1}{s} = 1 - \frac{1}{r},$$

so

$$\frac{1}{s} + \frac{1}{r} = 1. \tag{23}$$

Thus s and r are conjugate exponents. Solving explicitly gives

$$s = \frac{r}{r-1} > 1,$$

because $r > 1$.

Step 2: verify that part (a) applies to a and b

Equation (22) has the same form as the hypothesis in part (a), with the smaller output exponent now called s:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{s}.$$

We also need $p > s$ and $q > s$. Since

$$\frac{1}{s} = \frac{1}{p} + \frac{1}{q} > \frac{1}{p},$$

and all numbers are positive, we have $s < p$. Similarly, $s < q$. Therefore part (a) gives

$$\left(\sum_{j=1}^{n} (a_j b_j)^s\right)^{1/s} \leq \left(\sum_{j=1}^{n} a_j^p\right)^{1/p} \left(\sum_{j=1}^{n} b_j^q\right)^{1/q}. \tag{24}$$

Step 3: pair the product $a_j b_j$ with c_j

Because $1/s + 1/r = 1$, ordinary Hölder applied to the two sequences

$$x_j = a_j b_j, \qquad y_j = c_j$$

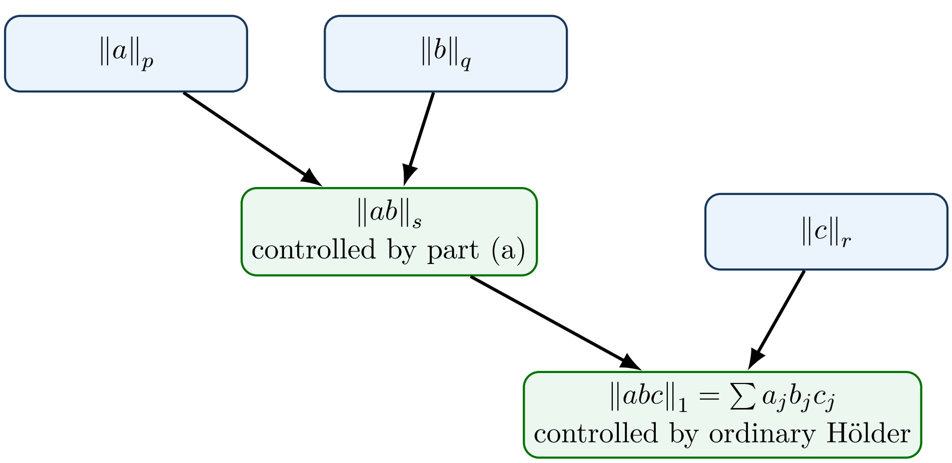
gives

$$\begin{aligned} \sum_{j=1}^{n} a_j b_j c_j &= \sum_{j=1}^{n} (a_j b_j) c_j \\ &\leq \left(\sum_{j=1}^{n} (a_j b_j)^s\right)^{1/s} \left(\sum_{j=1}^{n} c_j^r\right)^{1/r}. \end{aligned} \tag{25}$$

Now substitute the estimate (24) into (25):

$$\sum_{j=1}^{n} a_j b_j c_j \leq \left(\sum_{j=1}^{n} a_j^p\right)^{1/p} \left(\sum_{j=1}^{n} b_j^q\right)^{1/q} \left(\sum_{j=1}^{n} c_j^r\right)^{1/r}.$$

This is exactly (B).



5.1 A concrete equality example

Choose

$$p = q = r = 3,$$

for which $1/3 + 1/3 + 1/3 = 1$. Let

$$a = b = c = (1, 2).$$

Then the left side is

$$1 \cdot 1 \cdot 1 + 2 \cdot 2 \cdot 2 = 1 + 8 = 9.$$

Each factor on the right equals

$$(1^3 + 2^3)^{1/3} = 9^{1/3}.$$

Therefore the right side is

$$9^{1/3} \cdot 9^{1/3} \cdot 9^{1/3} = 9.$$

Again, equality occurs.

5.2 Equality in the triple-product inequality

A convenient way to state the equality condition is that the three powered sequences have the same shape. More precisely, equality occurs when there is a nonnegative sequence (d_j) and constants $\alpha, \beta, \gamma > 0$ such that

$$a_j^p = \alpha d_j, \qquad b_j^q = \beta d_j, \qquad c_j^r = \gamma d_j \qquad \text{for every } j. \tag{26}$$

For $p = q = r = 3$ and $a = b = c$, this condition is automatic.

6 Why this is called self-generalization

Ordinary Hölder looks like a two-sequence theorem with a reciprocal sum equal to 1. Part (a) shows that it automatically creates a whole family of two-sequence inequalities:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r} \quad \Longrightarrow \quad \|ab\|_r \le \|a\|_p \|b\|_q \,. \tag{27}$$

Part (b) then uses that family to create a three-sequence inequality:

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1 \quad \Longrightarrow \quad \|abc\|_1 \le \|a\|_p \|b\|_q \|c\|_r \,. \tag{28}$$

The general reciprocal-addition rule

The same idea can be repeated. If positive exponents satisfy

$$\frac{1}{t} = \frac{1}{p_1} + \frac{1}{p_2} + \cdots + \frac{1}{p_k},$$

then, under the usual positivity conditions on the transformed Hölder exponents,

$$\left(\sum_{j=1}^{n} \left| x_j^{(1)} x_j^{(2)} \cdots x_j^{(k)} \right|^t \right)^{1/t} \le \prod_{i=1}^{k} \left(\sum_{j=1}^{n} \left| x_j^{(i)} \right|^{p_i} \right)^{1/p_i} . \tag{29}$$

The slogan is:

$$\boxed{\text{pointwise products make reciprocal exponents add.}}$$

For part (b), take $k = 3$ and $t = 1$. The condition in (29) becomes exactly

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1.$$

7 Common mistakes and how to avoid them

Mistake 1: choosing $P = r/p$ instead of $P = p/r$

We need $1/P = r/p$, so the correct choice is

$$P = \frac{p}{r}.$$

A quick check is that P must be greater than 1; since $p > r$, $p/r > 1$, while $r/p < 1$.

Mistake 2: applying ordinary Hölder directly with p, q in part (a)

Ordinary Hölder requires $1/p + 1/q = 1$, but part (a) gives $1/p + 1/q = 1/r$. The cure is to apply Hölder to a_j^r, b_j^r with exponents $p/r, q/r$.

Mistake 3: taking the r-th root incorrectly

From

$$X \le A^{r/p} B^{r/q},$$

we get

$$X^{1/r} \le A^{1/p} B^{1/q},$$

because exponents multiply: $(r/p)(1/r) = 1/p$.

Mistake 4: reusing the letter r for two jobs in part (b)

The exercise already uses r as the exponent on c_j. Introduce a new letter, such as s, for the exponent satisfying

$$\frac{1}{s} = \frac{1}{p} + \frac{1}{q}.$$

This keeps the two stages of the proof separate.

Mistake 5: forgetting absolute values for real sequences

For arbitrary real entries, use $|a_j|^p$ and $|a_j b_j|^r$. In part (b), begin with

$$\left|\sum a_j b_j c_j\right| \le \sum |a_j b_j c_j|.$$

8 A compact homework-ready proof

Part (a)

Let

$$P = \frac{p}{r}, \qquad Q = \frac{q}{r}.$$

Then $P, Q > 1$, and

$$\frac{1}{P} + \frac{1}{Q} = \frac{r}{p} + \frac{r}{q} = r\left(\frac{1}{p} + \frac{1}{q}\right) = 1.$$

Applying ordinary Hölder to the sequences (a_j^r) and (b_j^r),

$$\begin{aligned}\sum_{j=1}^{n}(a_jb_j)^r &\le \left(\sum_{j=1}^{n}(a_j^r)^{p/r}\right)^{r/p}\left(\sum_{j=1}^{n}(b_j^r)^{q/r}\right)^{r/q}\\ &= \left(\sum_{j=1}^{n}a_j^p\right)^{r/p}\left(\sum_{j=1}^{n}b_j^q\right)^{r/q}.\end{aligned}$$

Taking the r-th root gives

$$\left(\sum_{j=1}^{n}(a_jb_j)^r\right)^{1/r} \le \left(\sum_{j=1}^{n}a_j^p\right)^{1/p}\left(\sum_{j=1}^{n}b_j^q\right)^{1/q}.$$

Part (b)

Define $s > 1$ by

$$\frac{1}{s} = \frac{1}{p} + \frac{1}{q}.$$

Then the given relation implies $1/s + 1/r = 1$. Ordinary Hölder applied to (a_jb_j) and (c_j) gives

$$\sum_{j=1}^{n}a_jb_jc_j \le \left(\sum_{j=1}^{n}(a_jb_j)^s\right)^{1/s}\left(\sum_{j=1}^{n}c_j^r\right)^{1/r}.$$

Since $1/p + 1/q = 1/s$, part (a) gives

$$\left(\sum_{j=1}^{n}(a_jb_j)^s\right)^{1/s} \le \left(\sum_{j=1}^{n}a_j^p\right)^{1/p}\left(\sum_{j=1}^{n}b_j^q\right)^{1/q}.$$

Combining the last two inequalities proves

$$\sum_{j=1}^{n}a_jb_jc_j \le \left(\sum_{j=1}^{n}a_j^p\right)^{1/p}\left(\sum_{j=1}^{n}b_j^q\right)^{1/q}\left(\sum_{j=1}^{n}c_j^r\right)^{1/r}.$$

9 Self-check questions

Try these before reading the answers.

1. If $p = 6$ and $q = 3$, find r from $1/p + 1/q = 1/r$. Then find the Hölder exponents $P = p/r$ and $Q = q/r$.
2. Why does $p > r$ imply $p/r > 1$? Why is this important?
3. In part (a), simplify
$$(a_j^r)^{p/r} \qquad \text{and} \qquad \left(X^{r/p}\right)^{1/r}.$$
4. Suppose $p = q = 4$ and $r = 2$ in part (b). Find the intermediate exponent s satisfying $1/s = 1/p+1/q$. Check that s and r are conjugate.
5. For $p = q = r = 3$ and $a = b = c = (1, 1)$, calculate both sides of the triple-product inequality.
6. Complete the slogan: pointwise products make ________________ add.

Answers

1. $$\frac{1}{r} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}, \qquad r = 2.$$
Thus
$$P = \frac{6}{2} = 3, \qquad Q = \frac{3}{2},$$
and $1/3 + 2/3 = 1$.
2. Dividing the positive inequality $p > r$ by r gives $p/r > 1$. Ordinary Hölder requires its exponents to be greater than 1.
3. $$(a_j^r)^{p/r} = a_j^p, \qquad (X^{r/p})^{1/r} = X^{1/p}.$$
4. $$\frac{1}{s} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \qquad s = 2.$$
Since $r = 2$, $1/s + 1/r = 1/2 + 1/2 = 1$.
5. The left side is $1 + 1 = 2$. Each factor on the right is $(1^3 + 1^3)^{1/3} = 2^{1/3}$, so their product is 2. Equality holds.
6. Reciprocal exponents.

10 Final summary

Part (a)

From
$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r},$$
define
$$P = \frac{p}{r}, \qquad Q = \frac{q}{r}.$$
Then $1/P + 1/Q = 1$. Apply ordinary Hölder to a_j^r and b_j^r, and finally take the r-th root.

Part (b)

From
$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1,$$
define s by
$$\frac{1}{s} = \frac{1}{p} + \frac{1}{q}.$$
Then $1/s + 1/r = 1$. First use part (a) to control $\|ab\|_s$, and then use ordinary Hölder to combine ab with c.

The essential lesson is not a difficult computation. It is the choice of exponents:

$$\boxed{\text{raise the sequences to suitable powers so that the new reciprocals add to 1.}}$$